

Nathan Buckner

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Staff Software Engineer with 17 years of experience designing frameworks, distributed systems, and developer tooling. Track record of creating widely-adopted internal tools from scratch - test frameworks, code generation pipelines, and platform abstractions. Strong collaborator who mentors teams and drives technical standards across organizations.

Professional Experience

Roku Inc. - Staff Software Engineer **Jul 2021 - Present**

- Architected multicloud service migration (AWS to GCP) - designed configuration abstraction layer enabling services to run across cloud providers without code changes
- Technical lead for payments platform serving millions of transactions - built payment routing engine with failover logic and provider-agnostic abstractions
- Scaled STAF testing framework into an org-wide ecosystem - designed plugin architecture, namespace mixins, declarative test configuration, and code generation from protobuf schemas
- Built infrastructure validation service (Python/FastAPI) to verify deployment correctness across distributed environments
- Designed protobuf schema registry with auto-generated typed clients in Python and Java, enforcing API contracts across 20+ microservices
- Led ECS-to-Kubernetes migration - container orchestration, service mesh (Istio), and GitOps deployment pipelines

Roku Inc. - Senior Software Engineer **Jun 2018 - Jul 2021**

- Created STAF - a Python test framework with decorator-based configuration, namespace resolution, and extensible plugin system for service integration testing
- Designed and implemented event-driven payment collection service processing provider webhook events at scale
- Built AWS EMR/Spark ETL pipelines transforming billing data for the data warehouse
- Designed payment gateway switch service with weighted routing, sticky sessions, and automatic failover
- Created reusable GitLab CI component library (YAML DSL) adopted across the engineering org
- Built protobuf-based service contract framework with code generation for typed Python/Java client stubs

Amazon Lab126 - Senior Software Developer L5,L6 **May 2017 - Jun 2018**

- Designed and built Burn - a Python test framework adopted cross-org, featuring declarative test configuration, composable mixins, and device abstraction layers
- Designed page object model with graph-based navigation for UI automation
- Mentored and trained local and overseas engineering teams on framework design patterns

Rackspace - Software Developer/Security Developer in Test II-IV **Mar 2012 - Feb 2017**

- Core developer of Syntribos - an open source security fuzzing tool accepted into the OpenStack community
- Designed and built OpenCafe - a Python test framework with composable behaviors and client abstractions; trained all new hires in onboarding class
- Built metrics API backed by object storage with ELK stack for data analysis and dashboarding
- Designed test harness for software defined networks with complex deployment orchestration
- Security work: threat models, code reviews, infrastructure/web app penetration testing

General Dynamics Mission Systems - Software Engineer **May 2009 - Jun 2012**

- Embedded systems development - kernel, POSIX, and application layers for military radio platforms
- Implemented RIPv2 routing protocol
- Secret Clearance

US Army - Signal Support System Specialist **May 2000 - Jun 2006**

- Network engineering (CCNA) and Linux/Windows system administration in tactical environments
- Secret Clearance, COMSEC custodian

Technical Expertise

Languages - Python (expert), Java, Shell/Bash, C (embedded), Terraform (HCL)

Systems - Distributed systems, service mesh (Istio), event-driven architecture, Kafka, Protobuf/Avro serialization, Linux internals

Data - Spark/EMR, DynamoDB, Spanner, Redis, Aerospike, SQL + ORMs

Infrastructure - AWS, GCP, Kubernetes, Docker, Terraform, GitOps (Helm/Helmfile)

Practices - Framework/DSL design, code generation, type-safe API contracts, CI/CD pipeline architecture, open source development

Education

University of Texas at San Antonio - Bachelor's in Computer Science **2006 - 2010**

Certifications - CCNA, Network+

Online Presence

GitHub - <https://github.com/bucknerns>

LinkedIn - <https://www.linkedin.com/in/nathanbuckner>